Anav Saxena

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EDUCATION

University of California, Los Angeles – B.S. in Mechanical Engineering Expected June 2029
Relevant Coursework – Statics, Advanced Strength of Materials, Dynamics, Science of Materials **GPA** – 3.904

EXPERIENCE

Wheels Responsible Engineer – Bruin Formula Racing, Formula SAE 2025 – present

- Own hub/upright assembly design to 74-lb suspension mass target, integrating Vehicle Dynamics hardpoints
- Perform Ansys FEA for max cornering/braking loads (0.392G long, 1.884G lat, 1.3 FS) to minimize unsprung mass
- Developed in-house upright machining plan projected to reduce subsystem manufacturing cost by ~$900 (23%)
- Selected shim camber tuning and deep-groove ball bearing design via trade studies weighing serviceability and cost
- Derived axial/radial bearing loads to select hub bearings and define relief groove dimensions based in ISO 18388
- Presented minimal viable product at team-wide design review, incorporating feedback to meet design objectives
- Designed custom electronics enclosures to meet reliability, waterproofing, and roadside accessibility requirements

Delta-Style Robotic Henna Applicator – Personal Project 2026 – present

- Design delta-style robot to dispense a viscoelastic, non-Newtonian fluid onto a hand on a 9” diameter build plate
- Develop hand-contour mapping on Klipper firmware for deposition on organic surfaces with constant Z-offset
- Integrate ISO 13850-based, hard-wired, normally-closed E-stop into Manta M5P/Raspberry Pi CM4 control system
- Architect frame/motion system, sizing NEMA 17 stepper motors, GT2 belts, and rails for precise nozzle kinematics

CNC Machining Mentor – Goldstrikers (FIRST Robotics Team 2473) Summer 2026 | Student: 2021 – 2025

- Validated Tormach 770M operation after facility relocation, integrating shop air supply and verifying ATC
- Built parametric SFM/IPT calculator for polycarbonate and aluminum, standardizing team’s feeds and speeds
- Prototyped single-setup tabbing workholding strategy for polycarbonate sheet routing to reduce production time
- Trained students on machine operation, CAM, and workholding to sustain team’s CNC manufacturing capability
- Authored SOP for machine operation, Onshape CAM procedures, and maintenance guidelines

Instructor – UCLA ENGR 1UW (Underwater Robotics) 2026 – present

- Instruct class of 20 undergraduates to prototype, design, fabricate, test, and iterate underwater robots
- Reworked lessons to emphasize technical decision-making and design for manufacturing and assembly (DFMA)

Technician – UCLA Engineering Makerspace 2026 – present

- Support daily Makerspace operations (~80k student uses annually), including training students to use equipment
- Develop presets and maintenance policies for new Bambu Lab P1S printers, prioritizing machine longevity

Destructive Testing Engineer – Bruin Racing Baja SAE 2025 – 2026

- Designed load-cell rig to determine one-way bearing failure load and establish vehicle safety margins
- Iterated rig for 3-axis CNC milling and machined bearing housing on Haas CNC mill within ±0.001” tolerance
- Performed FEA and hand-calculated key stress to validate bearings as designed failure mode before testing

Carbon I-Beam Engineer – SAMPE (UCLA Materials Research Society) 2025 – 2026

- Performed composite FEA in Ansys to optimize carbon fiber I-beam geometry for 7,200-lbf load case
- Fabricated prepreg carbon-fiber layups, cured, and assembled beam for three-point bend testing
- Root-caused beam failure at 5,300-lbf to delamination near foam core and revised layup process for future beams

SKILLS

- **Software** – SOLIDWORKS, Ansys Mechanical (structural analysis), Onshape, Python, MATLAB
- **Manufacturing** – FDM, manual mill/lathe, Haas/Tormach CNC mill, Fusion 360/Onshape CAM, laser cutter, fixturing